Assignment 5: 2D Lagrangian Dynamics

16-384: Robot Kinematics and Dynamics

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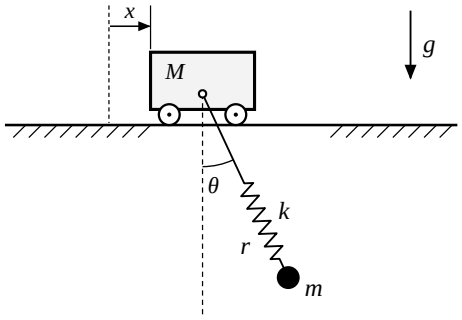
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1 Theory Questions

1) Spring, cart, pendulum *30 points*

The following spring-cart-pendulum system shown below is a modification of the pendulum-cart discussed in class. The cart has mass M, the pendulum has mass m and its spring has rest length r_0 and spring constant k.



a. [10 points] Write the potential and kinetic energy of the system in terms of the generalized coordinates $q = [x, r, \theta]^\top$ and their time derivatives $\dot{q} = [\dot{x}, \dot{r}, \dot{\theta}]^\top$. Then write the Lagrangian of the system.

$$T = \tfrac{1}{2}M\dot{x}^2 + \tfrac{1}{2}m[\dot{x}^2 + \dot{r}^2 + r^2\dot{\theta}^2 + 2\dot{x}\dot{r}\sin\theta + 2\dot{x}r\dot{\theta}\cos\theta]$$

$$V = -mgr\cos\theta + \tfrac{1}{2}k(r - r_0)^2$$

$$L = \tfrac{1}{2}M\dot{x}^2 + \tfrac{1}{2}m[\dot{x}^2 + \dot{r}^2 + r^2\dot{\theta}^2 + 2\dot{x}\dot{r}\sin\theta + 2\dot{x}r\dot{\theta}\cos\dot{\theta}] + mgr\cos\theta - \tfrac{1}{2}k(r - r_0)^2$$

b. [10 points] Write the derivatives of $\mathcal{L}$ with respect to q and $\dot{q}$ (6 in total).

$$\frac{\partial L}{\partial x} = 0$$

$$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 + m\dot{x}\dot{\theta}\cos\theta + mg\cos\theta - k(r - r_0)$$

$$\frac{\partial L}{\partial \theta} = m\dot{x}\dot{r}\cos\theta - m\dot{x}r\dot{\theta}\sin\theta - mgr\sin\theta$$

$$\frac{\partial L}{\partial \dot{x}} = (M + m)\dot{x} + m\dot{r}\sin\theta + mr\dot{\theta}\cos\theta$$

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r} + m\dot{x}\sin\theta$$

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} + m\dot{x}r\cos\theta$$

c. [5 points] Write the equations of motion of the system using the Euler-Lagrange equations.

$$(M+m)\ddot{x} + m\ddot{r}\sin\theta + m\dot{r}\ddot{\theta}\cos\theta = 0$$

$$m\ddot{r} + m\ddot{x}\sin\theta - mr\dot{\theta}^2 - m\dot{x}\dot{\theta}\cos\theta - mg\cos\theta + k(r - r_0) = 0$$

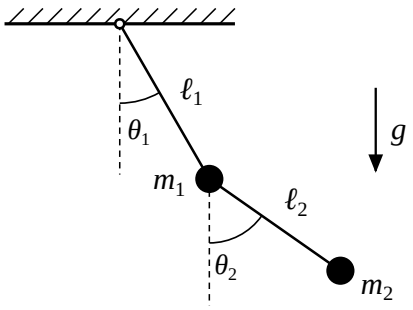
$$2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta} + m\ddot{x}r\cos\theta + m\dot{x}\dot{r}\cos\theta - m\dot{x}r\theta\sin\theta - m\dot{x}\dot{r}\cos\theta + m\dot{x}r\dot{\theta}\sin\theta + mgr\sin\theta = 0$$

d. [5 points] Suppose you can control the cart with an external force f_x. Write the equations of motion with the control input f_x.

2) Double Pendulum *30 points*

The double pendulum is a classic example of a chaotic system, yet its equations of motion are relatively simple to derive. You will be using your derived equations of motion to simulate the double pendulum in the coding section of this assignment.

The masses m_1 and m_2 are point masses at the ends of massless rods of length ℓ_1 and ℓ_2. Both angles are measured from the downward vertical, as shown in the figure: θ_2 is the absolute angle of link 2, *not* its angle relative to link 1. Gravity points in the $-y$ direction.



a. [10 points] Write the potential and kinetic energy of the system in terms of the generalized coordinates $q = [\theta_1, \theta_2]^\top$ and their time derivatives $\dot{q} = [\dot{\theta}_1, \dot{\theta}_2]^\top$.

$T = \frac{1}{2}m_1 l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2 cos(\theta_1 - \theta_2))$

$V = -m_1 g l_1 \cos\theta_1 - m_2 g(l_1\cos\theta_1 + l_2 cos\theta_2)$

$L = \frac{1}{2}m_1 l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2 cos(\theta_1 - \theta_2)) + m_1 g l_1 \cos\theta_1 + m_2 g(l_1\cos\theta_1 + l_2 cos\theta_2)$

b. [10 points] Write the derivatives of $\mathcal{L}$ with respect to q and $\dot{q}$ (4 in total).

$\frac{\partial L}{\partial \theta_1} = -m_2 l_1 l_2 \dot{\theta}_1\dot{\theta}_2 \sin(\theta_1 - \theta_2) - m_1 g l_1 \sin\theta_1 - m_2 g l_1 \sin\theta_1$

$\frac{\partial L}{\partial \theta_2} = m_2 g l_1 l_2 \dot{\theta}_1\dot{\theta}_2 \sin(\theta_1 - \theta_2) - m_2 g l_2 \sin\theta_2$

$\frac{\partial L}{\partial \dot{\theta}_1} = m_1 l_1^2\dot{\theta}_1 + m_2 l_1^2\dot{\theta}_1 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)$

$\frac{\partial L}{\partial \dot{\theta}_2} = m_2 l_2^2\dot{\theta}_2 + m_2 l_1 l_2 \dot{\theta}_1 \cos(\theta_1 - \theta_2)$

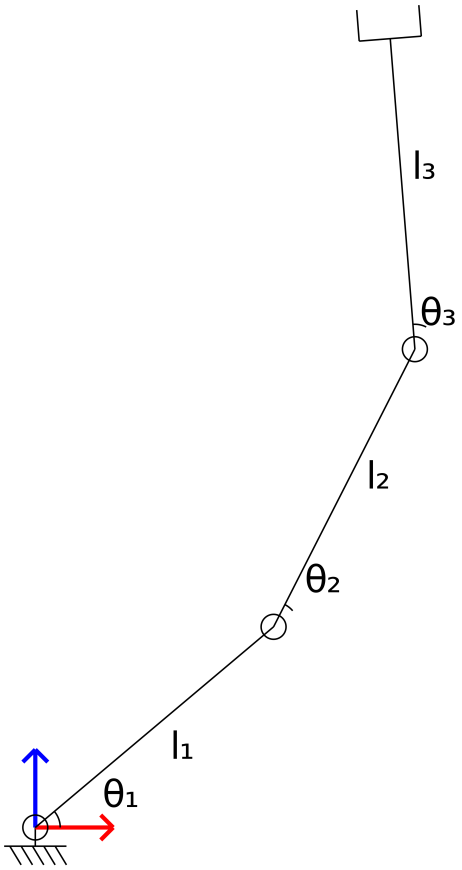
c. [10 points] Write the equations of motion of the system using the Euler-Lagrange equations.

$$m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + m_2 l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + m_1 g l_1 \sin\theta_1 + m_2 g l_1 \sin\theta_1 = 0$$

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + m_2 g l_2 \sin\theta_2 = 0$$

3) RRR Arm Dynamics *40 points*

Consider the following robot. Assume no joint limits, and that the links lengths are $l_1 = l_2 = l_3 = 1$, and the masses are $m_1 = m_2 = m_3 = 1$. The joint angles are measured as a relative angle from the previous link / horizontal, so $\theta_1 = \theta_2 = \theta_3 = 0$ means the arm is fully extended to the right. Each link's mass is spread uniformly along its length, so its center of mass is at its midpoint and its moment of inertia about its center of mass is $I_i = \frac{1}{12} m_i l_i^2 = \frac{1}{12}$. Gravity points in the $-y$ direction.



a. [10 points] Write the potential and kinetic energy of the system in terms of the generalized coordinates $q = [\theta_1, \theta_2, \theta_3]^\top$ and their time derivatives $\dot{q} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^\top$.

assumption: $I = \frac{1}{12}$ for RR arm, completing this for RR arm as recommended in piazza post

$T = \frac{1}{2}(m_1(v_1 \cdot v_1) + I_1 w_1^2) + \frac{1}{2}(m_2(v_2 \cdot v_2) + I_2 w_2^2)$

$T = \frac{2}{3}\dot{\theta}_1^2 + \frac{1}{8}(\dot{\theta}_1 + \dot{\theta}_2)^2 + \frac{1}{2}\dot{\theta}_1(\dot{\theta}_1 + \dot{\theta}_2)\cos\theta_2 + \frac{1}{24}(\dot{\theta}_1 + \dot{\theta}_2)^2$

$V = V_1 + V_2$

$V = \frac{1}{2}\sin\theta_1 + g(\sin\theta_1 + \frac{1}{2}\sin(\theta_1 + \theta_2)$

b. [10 points] Write the derivatives of $\mathcal{L}$ with respect to q and $\dot{q}$ (6 in total).

$\mathcal{L} = T_1 + T_2 - (V_1 + V_2)$

$\mathcal{L} = \frac{2}{3}\dot{\theta}_1^2 + \frac{1}{8}(\dot{\theta}_1 + \dot{\theta}_2)^2 + \frac{1}{2}\dot{\theta}_1(\dot{\theta}_1 + \dot{\theta}_2)\cos\theta_2 + \frac{1}{24}(\dot{\theta}_1 + \dot{\theta}_2)^2 - \frac{1}{2}\sin\theta_1 - g(\sin\theta_1 + \frac{1}{2}\sin(\theta_1 + \theta_2))$

$\frac{\partial L}{\partial \theta_1} = -\frac{1}{2}\cos\theta_1 - g\cos\theta_1 + \frac{1}{2}\cos(\theta_1 + \theta_2)$

$\frac{\partial L}{\partial \theta_2} = \frac{1}{2}\cos(\theta_1 + \theta_2) - \frac{1}{2}\dot{\theta}_1^2\sin\theta_1 - \frac{1}{2}\dot{\theta}_1\dot{\theta}_2\sin\theta_2$

$\frac{\partial L}{\partial \dot{\theta}_1} = \frac{4}{3}\dot{\theta}_1 + \frac{1}{4}(\dot{\theta}_1 + \dot{\theta}_2) + \dot{\theta}_1\cos\theta_2 + \frac{1}{2}\dot{\theta}_2\cos\theta_2 + \frac{1}{12}(\dot{\theta}_1 + \dot{\theta}_2)$

$\frac{\partial L}{\partial \dot{\theta}_2} = \frac{1}{4}(\dot{\theta}_1 + \dot{\theta}_2) + \frac{1}{2}\dot{\theta}_1\cos\theta_2 + \frac{1}{12}(\dot{\theta}_1 + \dot{\theta}_2)$

c. [10 points] Write the equations of motion of the system using the Euler-Lagrange equations. Include external torques τ_1, τ_2, τ_3 applied at each joint.

$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\theta}_1}) - \frac{\partial L}{\partial \theta_1} = \tau_1$

$\frac{4}{3}\ddot{\theta}_1 + \frac{1}{4}(\ddot{\theta}_1 + \ddot{\theta}_2) + \ddot{\theta}_1\cos\theta_2 - \dot{\theta}_1\dot{\theta}_2\sin\theta_2 + \frac{1}{2}\ddot{\theta}_2\cos\theta_2 - \frac{1}{2}\dot{\theta}_2^2\sin\theta_2 + \frac{1}{12}(\ddot{\theta}_1 + \ddot{\theta}_2) - \frac{1}{2}\cos\theta_1 + g\cos\theta_1 - \frac{1}{2}\cos(\theta_1 + \theta_2) = \tau_1$

$\tau_1 = \left(\frac{5}{3} + \cos\theta_2\right)\ddot{\theta}_1 + \left(\frac{1}{3} + \frac{1}{2}\cos\theta_2\right)\ddot{\theta}_2 - \dot{\theta}_1\dot{\theta}_2\sin\theta_2 - \frac{1}{2}\dot{\theta}_2^2\sin\theta_2 + \left(g - \frac{1}{2}\right)\cos\theta_1 - \frac{1}{2}\cos(\theta_1 + \theta_2)$

$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\theta}_2}) - \frac{\partial L}{\partial \theta_2} = \tau_2$

$\frac{1}{4}(\ddot{\theta}_1 + \ddot{\theta}_2) + \frac{1}{2}\ddot{\theta}_1\cos\theta_2 - \frac{1}{2}\dot{\theta}_1\dot{\theta}_2\sin\theta_2 + \frac{1}{12}(\ddot{\theta}_1 + \ddot{\theta}_2) - \frac{1}{2}\cos(\theta_1 + \theta_2) + \frac{1}{2}\dot{\theta}_1^2\sin\theta_1 + \frac{1}{2}\dot{\theta}_1\dot{\theta}_2\sin\theta_2 = \tau_2$

$\tau_2 = \left(\frac{1}{3} + \frac{1}{2}\cos\theta_2\right)\ddot{\theta}_1 + \frac{1}{3}\ddot{\theta}_2 + \frac{1}{2}\dot{\theta}_1^2\sin\theta_1 - \frac{1}{2}\cos(\theta_1 + \theta_2)$

d. [5 points] Find the standard form matrices $M(q)$, $C(q, \dot{q})$, and $G(q)$ such that the equations of motion can be written as $M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$.

$$M(q) = \begin{bmatrix} \frac{5}{3} + \cos\theta_2 & \frac{1}{3} + \frac{1}{2}\cos\theta_2 \\ \frac{1}{3} + \frac{1}{2}\cos\theta_2 & \frac{1}{3} \end{bmatrix}$$
$$C(q, \dot{q}) = \begin{bmatrix} 0 & -\frac{1}{2}(2\dot{\theta}_1 + \dot{\theta}_2)\sin\theta_2 \\ \frac{1}{2}\dot{\theta}_1 \sin\theta_1 & 0 \end{bmatrix}$$
$$G(q) = \begin{bmatrix} (g - \frac{1}{2})\cos\theta_1 - \frac{1}{2}\cos(\theta_1 + \theta_2) \\ -\frac{1}{2}\cos(\theta_1 + \theta_2) \end{bmatrix}$$

e. [5 points] What are the joint torques τ_1, τ_2, τ_3 required to hold the arm in the configuration $\theta_1 = \theta_2 = \theta_3 = 0$ (fully extended to the right) with no motion? How does this compare to the Jacobian-based gravity compensation method discussed previously?

$$\tau_1 = M(q)(0) + C(q, \dot{q})(0) + G(q)$$

$$G(0) = \begin{bmatrix} (g - \frac{1}{2})\cos(0) - \frac{1}{2}\cos(0) \\ -\frac{1}{2}\cos(0) \end{bmatrix}$$
$$\tau_1 = g - 1$$

$$\tau_1 = -\frac{1}{2}$$

Both approaches would yield the same result for τ_1 and τ_2

2 Submission Checklist

- ☐ Create a PDF of your answers to the written questions with the name `<andrew_id>.pdf` and submit it to Gradescope.